Unit Test Report

Project: Railway Reservation System (RRS), Team 13 Jackfruit Phase-1
Generated: 12-Nov-2026
Environment: Python 3.11.9, Windows
Tool: pytest

Summary

Total	Passed	Failed	Errors	Skipped	Pass rate	Run time
231	231	0	0	0	100.0%	12.1s

Results by test file

Test file	Tests	Passed	Failed	Skipped
test_admin.py	11	11	0	0
test_admin_routes.py	10	10	0	0
test_audit_repo.py	9	9	0	0
test_auth.py	6	6	0	0
test_availability.py	5	5	0	0
test_booking.py	14	14	0	0
test_booking_repo.py	1	1	0	0
test_cancellation.py	8	8	0	0
test_lockout.py	8	8	0	0
test_monitoring.py	13	13	0	0
test_payment.py	15	15	0	0
test_payment_repo.py	9	9	0	0
test_payment_security.py	1	1	0	0
test_performance.py	2	2	0	0
test_pnr.py	10	10	0	0
test_pnr_routes.py	6	6	0	0
test_promotion.py	16	16	0	0
test_rbac_audit.py	11	11	0	0
test_refund.py	8	8	0	0
test_search.py	6	6	0	0

test_seat_allocation.py	7	7	0	0
test_sensitive_data.py	7	7	0	0
test_session_timeout.py	7	7	0	0
test_train_admin.py	14	14	0	0
test_train_admin_routes.py	10	10	0	0
test_transport_security.py	7	7	0	0
test_usability.py	10	10	0	0

Failures

None. All executed tests passed.

Every test

test_admin.py

Test	Result	Time (s)
test_parse_date_range_valid	passed	0.008
test_parse_date_range_optional	passed	0.001
test_parse_date_range_invalid_format	passed	0.001
test_parse_date_range_start_after_end	passed	0.001
test_generate_report_calls_repository	passed	0.002
test_generate_report_rejects_invalid_date_range	passed	0.002
test_get_booking_report_aggregation	passed	0.010
test_get_occupancy_report_aggregation	passed	0.007
test_get_revenue_report_aggregation	passed	0.007
test_empty_reports_handle_zeroes_gracefully	passed	0.007
test_get_audit_trail_validation	passed	0.001

test_admin_routes.py

Test	Result	Time (s)
test_admin_reports_page_requires_login	passed	0.019
test_admin_reports_page_rejects_passenger	passed	0.034
test_admin_reports_page_accessible_by_admin	passed	0.031
test_admin_reports_api_requires_login	passed	0.002

test_admin_reports_api_rejects_passenger	passed	0.004
test_admin_reports_api_accessible_by_admin	passed	0.005
test_admin_audit_api_rejects_passenger	passed	0.006
test_admin_audit_api_accessible_by_admin	passed	0.004
test_create_booking_api_success	passed	0.008
test_create_booking_api_validation_error	passed	0.005

test_audit_repo.py

Test	Result	Time (s)
test_write_audit_log	passed	0.005
test_get_booking_audit_logs	passed	0.004
test_all_six_required_lifecycle_actions_supported[BOOKING_CREATED]	passed	0.004
test_all_six_required_lifecycle_actions_supported[PAYMENT_SUCCESS]	passed	0.003
test_all_six_required_lifecycle_actions_supported[PAYMENT_FAILED]	passed	0.003
test_all_six_required_lifecycle_actions_supported[PAYMENT_TIMEOUT]	passed	0.004
test_all_six_required_lifecycle_actions_supported[BOOKING_CANCELLED]	passed	0.003
test_all_six_required_lifecycle_actions_supported[BOOKING_EXPIRED]	passed	0.003
test_get_recent_audit_logs_unfiltered_and_filtered	passed	0.006

test_auth.py

Test	Result	Time (s)
test_valid_registration_creates_account	passed	0.143
test_duplicate_email_rejected	passed	0.299
test_invalid_registration_fields_rejected	passed	0.001
test_password_is_hashed_not_plaintext	passed	0.145
test_login_with_correct_credentials	passed	0.299
test_login_with_wrong_password_or_unknown_user	passed	0.290

test_availability.py

Test	Result	Time (s)
test_available_seats	passed	0.002
test_fully_booked	passed	0.002

test_invalid_train_id	passed	0.002
test_invalid_class	passed	0.001
test_train_not_found	passed	0.008

test_booking.py

Test	Result	Time (s)
test_valid_booking_creates_provisional_booking	passed	0.002
test_invalid_booking_details_are_rejected[user_id-0]	passed	0.002
test_invalid_booking_details_are_rejected[schedule_id-0]	passed	0.002
test_invalid_booking_details_are_rejected[passenger_name-]	passed	0.002
test_invalid_booking_details_are_rejected[passenger_name-]	passed	0.003
test_invalid_booking_details_are_rejected[passenger_age-abc]	passed	0.002
test_invalid_booking_details_are_rejected[passenger_age-0]	passed	0.001
test_invalid_booking_details_are_rejected[passenger_age-121]	passed	0.002
test_invalid_booking_details_are_rejected[passenger_gender-]	passed	0.001
test_invalid_booking_details_are_rejected[passenger_gender-Unknown]	passed	0.002
test_invalid_booking_details_are_rejected[class_code-1A]	passed	0.001
test_invalid_booking_details_are_rejected[berth_preference-Invalid]	passed	0.001
test_passenger_name_is_trimmed	passed	0.002
test_optional_berth_preference_can_be_empty	passed	0.001

test_booking_repo.py

Test	Result	Time (s)
test_expired_bookings_generate_audit_events	passed	0.008

test_cancellation.py

Test	Result	Time (s)
test_cancel_updates_status_and_records_refund	passed	0.001
test_refund_shrinks_close_to_departure	passed	0.001
test_preview_does_not_cancel	passed	0.001
test_confirmation_is_required	passed	0.001
test_cannot_cancel_someone_elses_ticket	passed	0.001

test_admin_can_cancel_any_ticket	passed	0.001
test_cannot_cancel_twice	passed	0.002
test_unknown_or_malformed_pnr	passed	0.001

test_lockout.py

Test	Result	Time (s)
test_four_failures_do_not_lock	passed	0.884
test_fifth_failure_locks_account	passed	0.895
test_locked_account_refuses_even_correct_password	passed	1.689
test_lock_ends_after_lockout_period	passed	1.821
test_success_resets_failure_counter	passed	1.485
test_unknown_email_never_raises_locked	passed	0.155
test_minutes_left_rounds_up	passed	0.001
test_naive_datetime_from_db_is_treated_as_utc	passed	0.147

test_monitoring.py

Test	Result	Time (s)
test_retry_succeeds_after_transient_failures_with_doubling_delay	passed	0.002
test_retry_gives_up_after_the_last_attempt	passed	0.001
test_retry_does_not_repeat_other_errors	passed	0.001
test_metrics_count_requests_and_server_errors	passed	0.001
test_empty_metrics_have_zero_error_rate	passed	0.002
test_availability_percentage_from_log	passed	0.083
test_availability_window_and_missing_or_damaged_log	passed	0.023
test_probe_once_reports_up_down_and_unreachable	passed	1.572
test_health_is_200_when_database_answers_and_503_otherwise_without_leaking	passed	0.014
test_db_ok_hides_exception_details	passed	0.009
test_availability_report_is_admin_only	passed	0.013
test_availability_report_contents	passed	0.080
test_availability_report_without_data_yet	passed	0.007

test_payment.py

Test	Result	Time (s)
test_successful_payment	passed	0.003
test_rac_booking_gets_no_pnr	passed	0.004
test_unsuccessful_payment[FAILED]	passed	0.003
test_unsuccessful_payment[TIMEOUT]	passed	0.002
test_transaction_reference_generated	passed	0.001
test_invalid_user_id[0]	passed	0.002
test_invalid_user_id[-1]	passed	0.002
test_invalid_user_id[1]	passed	0.002
test_invalid_user_id[True]	passed	0.002
test_invalid_booking_id[0]	passed	0.001
test_invalid_booking_id[-1]	passed	0.002
test_invalid_booking_id[1]	passed	0.002
test_invalid_booking_id[True]	passed	0.003
test_invalid_payment_outcome	passed	0.002
test_payment_outcome_is_case_insensitive	passed	0.004

test_payment_repo.py

Test	Result	Time (s)
test_successful_payment_confirms_booking_and_records_audit	passed	0.007
test_failed_payment_releases_seat_and_records_audit	passed	0.006
test_timeout_payment_releases_seat_and_records_audit	passed	0.006
test_expired_hold_forces_timeout_status	passed	0.013
test_duplicate_payment_rejected	passed	0.017
test_payment_rejected_when_not_provisional	passed	0.013
test_payment_rejected_when_booking_not_found	passed	0.012
test_invalid_status_and_txn_ref_rejected	passed	0.002
test_get_booking_for_user	passed	0.007

test_payment_security.py

Test	Result	Time (s)
test_payments_table_does_not_store_card_details	passed	0.004

test_performance.py

Test	Result	Time (s)
test_booking_transaction_performance_90th_percentile	passed	0.137
test_payment_failure_workflow_performance	passed	0.004

test_pnr.py

Test	Result	Time (s)
test_generated_pnr_is_ten_digits	passed	0.001
test_generate_retries_until_unique	passed	0.001
test_generate_gives_up_after_attempts	passed	0.001
test_confirmed_booking_gets_a_pnr	passed	0.003
test_assign_is_idempotent	passed	0.001
test_unconfirmed_or_missing_booking_gets_no_pnr	passed	0.002
test_lookup_returns_correct_status	passed	0.001
test_lookup_unknown_pnr_returns_none	passed	0.002
test_lookup_rejects_malformed_pnr	passed	0.001
test_only_owner_or_admin_can_look_up	passed	0.001

test_pnr_routes.py

Test	Result	Time (s)
test_api_requires_login	passed	0.013
test_pnr_status_hides_internal_user_id	passed	0.010
test_pnr_status_not_found_and_bad_input	passed	0.018
test_cancel_success_and_errors	passed	0.033
test_cancel_without_confirm_flag_is_rejected_by_service	passed	0.009
test_allocation_status_endpoint	passed	0.025

test_promotion.py

Test	Result	Time (s)
test_free_seat_goes_to_earliest_rac_then_waitlist_refills_rac	passed	0.001
test_nothing_changes_when_no_seat_is_free_and_rac_is_full	passed	0.001

test_booking_order_is_respected	passed	0.001
test_waitlisted_gets_seat_directly_when_nobody_is_in_rac	passed	0.001
test_two_free_seats_promote_two_rac_passengers	passed	0.001
test_rac_cancellation_pulls_waitlist_into_rac	passed	0.002
test_rac_quota_is_never_exceeded	passed	0.001
test_empty_queues_and_overbooked_inputs_are_safe	passed	0.002
test_plan_is_idempotent_once_applied	passed	0.001
test_rebalance_after_cancel_hands_planner_and_pnr_maker_to_repo	passed	0.001
test_repo_promotes_rac_to_confirmed_with_pnr_and_waitlist_to_rac	passed	0.017
test_repo_second_run_promotes_nobody_again	passed	0.005
test_repo_booking_order_uses_created_at_then_id	passed	0.002
test_repo_unknown_booking_or_no_free_seat_changes_nothing	passed	0.003
test_repo_unexpired_provisional_hold_keeps_the_seat	passed	0.002
test_cancel_booking_calls_rebalance_and_survives_its_failure	passed	0.011

test_rbac_audit.py

Test	Result	Time (s)
test_anonymous_gets_401_on_admin_api	passed	0.010
test_passenger_direct_api_call_gets_403	passed	0.009
test_admin_is_allowed	passed	0.009
test_unknown_or_blank_role_is_refused	passed	0.039
test_html_admin_page_never_leaks_to_passenger	passed	0.021
test_audit_log_endpoint_is_admin_only	passed	0.016
test_log_action_records_user_action_and_details	passed	0.001
test_log_action_requires_user_and_action	passed	0.001
test_details_are_redacted	passed	0.001
test_recent_entries_newest_first_and_limit_clamped	passed	0.001
test_audit_log_endpoint_returns_entries_to_admin	passed	0.022

test_refund.py

Test	Result	Time (s)
test_percent_by_time_left	passed	0.001

test_threshold_boundaries_are_inclusive	passed	0.002
test_no_refund_after_departure	passed	0.001
test_default_policy_used_when_table_empty	passed	0.001
test_amount_matches_policy	passed	0.001
test_amount_is_rounded_to_paise	passed	0.001
test_zero_refund_amount	passed	0.001
test_hours_before_departure	passed	0.003

test_search.py

Test	Result	Time (s)
test_matching_trains_returned_with_class_and_timing	passed	0.001
test_no_match_returns_empty_list	passed	0.001
test_missing_fields_rejected_without_querying	passed	0.003
test_same_source_and_destination_rejected	passed	0.001
test_past_and_malformed_dates_rejected	passed	0.002
test_today_is_allowed	passed	0.001

test_seat_allocation.py

Test	Result	Time (s)
test_rac_capacity_is_ten_percent_rounded_up	passed	0.002
test_confirmed_while_seats_are_free	passed	0.001
test_rac_when_full_and_rac_quota_open	passed	0.001
test_waitlisted_when_full_and_rac_quota_full	passed	0.001
test_assign_status_uses_repo_counts	passed	0.001
test_assign_status_unknown_schedule_returns_none	passed	0.001
test_assign_status_validates_input	passed	0.001

test_sensitive_data.py

Test	Result	Time (s)
test_users_table_stores_only_a_password_hash	passed	0.002
test_no_table_has_card_or_secret_columns	passed	0.003
test_registration_hashes_with_a_salted_algorithm	passed	0.297

test_session_cookie_never_contains_password_or_hash	passed	0.006
test_app_login_route_stores_only_safe_keys	passed	0.015
test_application_code_never_logs_credentials	passed	0.025
test_session_cookie_flags_are_hardened	passed	0.001

test_session_timeout.py

Test	Result	Time (s)
test_is_expired_boundaries	passed	0.001
test_active_session_passes_and_is_refreshed	passed	0.010
test_idle_api_call_gets_401_and_session_is_cleared	passed	0.008
test_old_token_stays_dead_after_expiry	passed	0.014
test_idle_page_request_redirects_to_login	passed	0.008
test_session_without_timestamp_is_stamped_then_times_out	passed	0.010
test_anonymous_requests_are_untouched	passed	0.006

test_train_admin.py

Test	Result	Time (s)
test_valid_train_is_cleaned	passed	0.002
test_train_rejects_bad_fields	passed	0.001
test_partial_update_checks_only_supplied_fields	passed	0.001
test_classes_validation	passed	0.001
test_travel_date_validation	passed	0.001
test_add_train_records_actor_and_returns_view	passed	0.001
test_invalid_train_never_reaches_the_repository	passed	0.001
test_duplicate_train_becomes_conflict	passed	0.002
test_update_and_remove_train_not_found	passed	0.001
test_update_train_passes_only_changed_fields_and_audit_details	passed	0.001
test_add_schedule_cleans_classes_and_date	passed	0.001
test_schedule_clash_and_missing_train	passed	0.001
test_remove_schedule_with_bookings_is_a_conflict	passed	0.001
test_list_trains_groups_rows_into_trains_schedules_classes	passed	0.001

test_train_admin_routes.py

Test	Result	Time (s)
test_every_endpoint_rejects_anonymous_and_passengers	passed	0.156
test_passenger_cannot_change_anything	passed	0.010
test_admin_adds_train	passed	0.026
test_validation_error_is_400_with_messages	passed	0.014
test_form_posts_are_refused_to_block_csrf	passed	0.017
test_duplicate_is_409_and_missing_is_404	passed	0.031
test_update_and_remove_train	passed	0.014
test_schedule_endpoints	passed	0.020
test_list_endpoint_returns_trains	passed	0.014
test_page_is_admin_only	passed	0.038

test_transport_security.py

Test	Result	Time (s)
test_plain_http_page_is_redirected_to_https	passed	0.005
test_plain_http_api_and_posts_are_rejected	passed	0.006
test_https_requests_work_and_get_hsts	passed	0.008
test_rule_is_off_by_default_for_local_development	passed	0.005
test_forwarded_proto_is_ignored_unless_proxy_is_trusted	passed	0.010
test_safe_headers_are_always_sent	passed	0.007
test_ssl_context_refuses_anything_below_tls_1_2	passed	0.289

test_usability.py

Test	Result	Time (s)
test_every_form_field_has_a_label	passed	0.021
test_images_have_alt_text	passed	0.017
test_base_page_has_language_main_landmark_and_skip_link	passed	0.001
test_messages_are_announced_to_screen_readers	passed	0.001
test_search_form_has_sensible_defaults	passed	0.002
test_no_configuration_needed_to_start	passed	0.001

test_contrast_helper_matches_known_values	passed	0.001
test_status_badge_colours_meet_aa_contrast	passed	0.001
test_every_status_badge_also_has_a_symbol	passed	0.002
test_main_page_colours_meet_aa_contrast	passed	0.001